

EXP-6: Work with Sniffers for Monitoring Network Communication (Wireshark)

Prerequisites:

- Basic understanding of TCP/IP, Ethernet, and network protocols (HTTP, DNS, TCP, UDP, ARP).
- Familiarity with command-line basics.
- Wireshark installed on the lab machines (latest stable release recommended).

Learning Objectives

1. Understand how packet sniffers capture and display network traffic.
2. Use Wireshark to capture live traffic on an interface.
3. Apply capture and display filters to isolate relevant packets.
4. Inspect packet contents: Ethernet, IP, TCP/UDP headers and application payloads.
5. Reconstruct TCP streams and HTTP sessions.
6. Use Wireshark statistics and tools to summarize traffic and identify anomalies.
7. Export and save capture files (.pcapng) and use tshark for command-line analysis.

Safety & Legal Notice

- Only capture traffic on networks and devices for which you have explicit permission. Capturing someone else's private traffic without consent may be illegal and unethical.
- In a lab environment, use isolated networks, virtual machines, or intentionally generated traffic.

Equipment & Software

- Desktop or laptop with Wireshark installed.
- Network connection (wired preferred for stable captures) or isolated virtual network.
- Optional: Two VMs (one as client, one as server) to generate controlled traffic.

Lab Setup Instructions

1. Ensure Wireshark is installed. Launch Wireshark with sufficient privileges to capture from network interfaces (on Linux use sudo or run as root; on Windows run as Administrator).
2. Identify the network interface to capture from (e.g., Ethernet, Wi-Fi, vboxnet0, vmnet1).
3. If using virtual machines, set their network adapter to "Host-only" or "NAT with port forwarding" depending on the exercise.

Exercise 1 — Basic Capture and Filters

Goal: Capture live traffic and use display filters to inspect packets.

Steps:

1. Open Wireshark and start capturing on the chosen interface.
2. Generate network activity from the host (open a web page, ping a remote host, perform DNS lookups).
3. Stop capture after ~30–60 seconds.

Tasks:

- Use display filters to find specific traffic types (try these in the filter bar):
 - icmp — show ICMP (ping) packets.
 - http — show HTTP requests/responses.
 - tcp.port == 80 — show TCP traffic on port 80.
 - ip.addr == 192.168.1.10 — show packets to/from an IP address.
- Save the capture as `cstlab_ex6_<yourname>.pcapng`.

Questions:

1. How many ICMP echo requests and replies were captured?
2. Which IP addresses were involved in your web browsing activity?

Exercise 2 — Capture Filters vs Display Filters

Goal: Learn the difference and apply capture filters to limit data.

Notes:

- Capture filters (set before starting capture) use libpcap syntax and limit what data is recorded.
- Display filters (applied after capture) filter what you see without changing the recorded file.

Steps:

1. In Wireshark, click the capture interface settings and set the capture filter to port 53 or port 80.
2. Start capture and perform a web browse and a DNS lookup (e.g., `nslookup www.example.com`).
3. Stop capture.

Tasks:

- Compare the file size and packet count to the full capture from Exercise 1.
- Try capture filter examples:
 - host 192.168.1.5 — capture only traffic to/from a host.
 - tcp and not port 22 — capture TCP traffic excluding SSH.

Questions:

1. What are the advantages and disadvantages of using capture filters?

Exercise 3 — Protocol Dissection and Field Analysis

Goal: Inspect common protocol fields and headers.

Steps:

1. Open a capture file (use your saved file or a provided sample .pcapng).
2. Select a TCP packet containing an HTTP request.
3. Expand the packet details pane to view:
 - Frame
 - Ethernet II (src/dst MAC)
 - Internet Protocol Version 4 (src/dst IP, TTL)
 - Transmission Control Protocol (src/dst ports, seq/ack numbers, flags)
 - Hypertext Transfer Protocol (request/response headers)

Tasks:

- Identify the TCP flags set in the SYN, SYN/ACK, and FIN packets.
- Note the HTTP Host and User-Agent headers for a captured request.

Questions:

1. How does the sequence and acknowledgment numbering work during the TCP handshake?

Exercise 4 — Follow TCP Stream & Reassemble

Goal: Reconstruct conversations and view payloads as readable text.

Steps:

1. Right-click an HTTP packet and select **Follow > TCP Stream**.
2. Observe the full request/response conversation in text.
3. Use the menu to switch view from ASCII to EBCDIC or raw if needed.

Tasks:

- Save the reconstructed stream to a file.
- Reconstruct an SMTP or FTP conversation (if available) and identify credentials or commands (only in lab-controlled environment).

Questions:

1. Which HTTP status codes did you observe? What do they indicate?

Exercise 5 — DNS and ARP Analysis

Goal: Inspect name resolution and layer-2 address translation.

Steps:

1. Filter for DNS traffic: `dns or udp.port == 53`.
2. Perform DNS lookups from the command line and watch the traffic.
3. Filter for ARP: `arp` and observe ARP requests and replies.

Tasks:

- Identify query types (A, AAAA, PTR, CNAME).
- For ARP, note the sender and target hardware/IP addresses.

Questions:

1. What is the difference between iterative and recursive DNS queries (as observed in packets)?

Exercise 6 — TLS/HTTPS Analysis (What You Can See)

Goal: Understand limits when traffic is encrypted and what metadata remains visible.

Steps:

1. Capture an HTTPS session (browse to <https://www.example.com>).
2. Use display filters: `tls` or `ssl`.

Observations:

- You will see TCP handshake and TLS handshake messages but not application payload.
- You can observe server certificates (ClientHello/ServerHello, certificate chain), SNI (Server Name Indication) if not encrypted by TLS 1.3 ESNI, and cipher suites.

Tasks:

- Inspect the TLS Client Hello and list the cipher suites offered.
- View the server certificate details (subject, issuer, validity period).

Questions:

1. Why can't Wireshark display HTTPS payload by default? How can you decrypt TLS traffic in Wireshark (lab-only, with keys)?

Exercise 7 — Using Wireshark Statistics and Tools

Goal: Use built-in statistics to summarize traffic and identify anomalies.

Steps & Tasks:

- **Statistics > Protocol Hierarchy:** Observe protocol breakdown and byte/packet counts.
- **Statistics > Conversations:** View top talkers (IP/IP, TCP, UDP).
- **Statistics > Endpoints:** See which endpoints exchanged the most traffic.
- **Statistics > IO Graphs:** Plot packet/byte rates over time; create graphs per protocol.
- **Analyze > Expert Information:** Review potential problems, warnings, malformed packets.

Questions:

1. Which IP address generated the most traffic in your capture?
2. Did the Expert Information flag anything suspicious?

Exercise 8 — Exporting, Scripting, and tshark

Goal: Save captures, export data, and perform command-line analysis.

Steps:

1. File > Export Specified Packets or File > Export Packet Dissections to CSV/PDML/TEXT.
2. Save the capture as exp6_<yourname>.pcapng.
3. Use tshark (Wireshark's CLI):
 - List interfaces: tshark -D
 - Capture to file: tshark -i 1 -w capture.pcapng -a duration:30
 - Read and filter: tshark -r capture.pcapng -Y "http.request" -T fields -e ip.src -e http.host -e http.request.uri

Tasks:

- Export all HTTP request URIs seen to a text file using tshark.

Challenge Task — Detecting a Simple ARP Spoofing Attempt (Controlled Lab)

Goal: Use Wireshark to identify signs of ARP spoofing.

Setup (lab-only, controlled):

- Configure a VM to run an ARP spoofing tool against another lab VM.
- Capture traffic between VMs.

What to look for:

- Multiple ARP Replies claiming the same IP but with different MAC addresses.
- Duplicate IP addresses seen in ARP responses.
- An unusual increase in ARP traffic.

Questions:

1. Which packets indicate possible ARP poisoning?
2. How would you mitigate ARP spoofing on a real network?

Lab Report Requirements

Submit a short report (1–2 pages) including:

1. Objectives and setup description.
2. Steps performed and filters used.
3. Key observations and screenshots (Wireshark window with important packets highlighted).
4. Answers to questions in each exercise.
5. lab6_<yourname>.pcapng attached.

Assessment / Grading Rubric (Suggested)

- Proper setup and capture: 20%
- Correct use of filters and analyses: 30%
- Accuracy of observations and answers: 30%
- Report quality and screenshots: 20%

Additional Resources & References

- Wireshark Official Documentation: <https://www.wireshark.org/docs/>
- "Practical Packet Analysis" by Chris Sanders
- Wireshark User Guide and Display Filter Reference

Instructor Notes & Tips

- Encourage students to use virtual labs or isolated networks when practicing packet captures.
- Emphasize privacy and legality—never capture traffic on networks you don't own or manage.
- Provide sample .pcapng files for students who cannot generate certain traffic types.